Composite Indicator for Detecting and Localizing Time-Varying RFI in SAR Raw Data

Hengrui Zhang and Ning Li, *Member, IEEE*

***Abstract*— Radio frequency interference (RFI) seriously affects the processing of synthetic aperture radar (SAR) data and the interpretation of images. The accuracy of RFI detection, which is the first step of RFI suppression, directly affects the follow-up work. However, the implementations of existing RFI detection methods mostly rely on the assumption that the flatness of the range spectrum will be destroyed by strong RFI, making these methods ineffective in the face of weak energy RFI. Furthermore, in these methods, the problem of accurately localizing the RFI is rarely considered, resulting in a considerable loss of imaging signals during subsequent RFI suppression. In this manuscript, a method of time-varying RFI detection and localization is proposed with two steps that are based on the flatness and symmetry of the SAR range spectrum. This method can effectively avoid missed detection and accurately find the location of RFI with weaker energy in an echo. The proposed method first detects the existence of RFI along the azimuth direction. Moreover, the proposed composite index is used. The position of the RFI is then found along the range direction, and the log ratio operator and eigenvalue decomposition (EVD) are used. Experimental results on both simulated and raw data obtained by airborne and spaceborne SAR validate the advantages of the suggested method in detecting and locating weak energy RFI.**

***Index Terms*— Flatness, radio frequency interference (RFI), RFI detection, symmetry, synthetic aperture radar (SAR).**

I. Introduction

A. Background

SYNTHETIC aperture radar (SAR) is an active microwave sensor that recreates an irreplaceable role in the domain of remote sensing through its special working mode and the advantages of high-resolution earth observation. However, the global electromagnetic spectrum has become increasingly congested with the rapid development of military and civilian radio communications [1]. As a broadband radar system, SAR shares the same spectrum with radio systems and radar equipment in the same band [2]. Therefore, it is easy to receive radio frequency interference (RFI) from other wireless equipment [3], [4]. When RFI is present in the echoes received by an SAR system, the impulse response of the SAR signals will be disturbed, resulting in a deterioration of the imaging quality. More intuitively, an RFI will produce foggy, bright streaky, or raindrop-like artifacts superimposed on the SAR imaging area, resulting in inaccurate spatial and radiometric measurements [5]. The resulting polarization and phase distortion will further hinder the operational applications of SAR images, such as agricultural surveys [6], land planning [7], and ecological environment monitoring [8].

RFI detection is an essential preceding step in the interference suppression process. When an RFI is accurately detected, it is extracted from the echo, and the reconstruction of the useful signal is completed. The detection accuracy directly affects the follow-up work [9], [10]. However, most existing methods for detecting RFIs rely on the presumption that RFIs will destroy the flatness of SAR imaging signals [11], [12]. When weak energy RFI exists in strong SAR echoes, these methods cannot effectively detect RFI-containing pulses. In addition, these methods rarely consider the problem of accurately locating RFI, which makes it difficult to effectively protect useful signals during subsequent RFI suppression. Therefore, studying the detection and location methods of RFIs with weak energy for SAR data is of great application significance.

B. Previous Work

As a meaningful research topic in the field of SAR signal processing, the accurate detection of RFIs is directly related to the effect of subsequent suppression. For decades, scholars and research teams have proposed various detection methods of RFIs in SAR data [13], [14], which can be generally split into time-domain and transform-domain methods for detecting RFI [15], [16].

The detection methods applied in the time domain can effectively handle RFI signals with sinusoidal-like waveforms. The eigen-subspace projection (ESP) method takes advantage of the uncorrelated properties of the echo signal and RFI to construct the echo signal as a Hankel matrix [17]. The matrix is then decomposed into valid imaging signals and RFI

Manuscript received 20 June 2023; revised 28 September 2023 and 20 November 2023; accepted 16 December 2023. Date of publication 19 December 2023; date of current version 29 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61871175 and in part by the Graduate Education Innovation and Quality Improvement Program of Henan University under Grant SYL20060144. *(Corresponding author: Ning Li.)*

Hengrui Zhang is with the School of Computer and Information Engineering, Henan University, Kaifeng 475004, China, and with the Henan Province Engineering Research Center of Spatial Information Processing, Kaifeng 475004, China, and with the Henan Key Laboratory of Big Data Analysis and Processing, Kaifeng 475004, China, and also with the School of Information and Communication Engineering, Beijing University of Posts and Telecommunications, Beijing 100190, China (e-mail: zhr_henu@henu.edu.cn).

Ning Li is with the School of Computer and Information Engineering, Henan University, Kaifeng 475004, China, and with the Henan Province Engineering Research Center of Spatial Information Processing, Kaifeng 475004, China, and also with the Henan Key Laboratory of Big Data Analysis and Processing, Kaifeng 475004, China (e-mail: hedalining@henu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3345151

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

subspaces by eigenvalue decomposition (EVD) to complete RFI detection and suppression. The implementation of the kurtosis method [18] is based on the fact that strong RFI destroys the flatness of SAR echoes. This method calculates the kurtosis of the SAR echo pulse by pulse and then sets the threshold to detect whether an RFI exists. However, these detection methods based on statistical features have the disadvantage of being unable to determine the location of an RFI. In 2021, a time-domain detector (TDD) that uses adjacent pulse change detection technology was proposed [19], which can detect the existence of pulsed RFIs (PRFIs) and locate their positions in an echo. This method can accurately and sufficiently detect strong RFIs. However, the detection effect is decreased in the case of weak energy RFIs.

The main principle of the RFI detection method in the transform domain is to distinguish the characteristic difference between SAR echoes and RFIs in different transform domains. This type of method has many advantages. For example, the complex convolution process can be realized by simple multiplication in the frequency domain. Moreover, various filtering operations are relatively straightforward to execute in the transform domain. The transform domains applied to RFI detection in SAR data mainly include frequency, time-frequency [20], wavelet, and other domains [21], [22]. The range spectrum method [23] proposed in 1998 for detecting RFI is based on the short-time Fourier transform (STFT) of the echo in the frequency domain and calculates its average amplitude. However, this method may fail when the SAR signal has strong backscatter. To more accurately detect RFIs, the detection method named averaging-range-spectrum (ARS) [24] was proposed in 2003. Specifically, based on the coherence of RFIs, the method detects the presence of RFIs after averaging a set of adjacent pulses in the frequency domain. The improved sub-band kurtosis method [25], which was proposed in 2013, combines the ARS and kurtosis methods. In this approach, the range spectrum is split into sub-bands using the holographic technology of spaceborne SAR imaging, and a second detection step is added on this basis. As a result, it is possible to quickly set the detection threshold while improving the detection probability of RFIs. In 2017, the local autocorrelation detection method [26] was proposed to detect intermittently transmitted wideband RFI effectively in the azimuth-frequency domain. In 2018, a detection and localization method of RFIs employing digital beamforming (DBF) was proposed [27]. In this method, RFIs are detected with a time bandpass filter, and the DBF is used to evaluate the angle of arrival to spatially locate the RFIs. In 2022, a time-frequency detector (TFD) method was proposed [28]. This two-step approach combines EVD and STFT to accumulate ridge energy in the time-frequency domain. Compared to TDD, TFD can detect PRFIs even when it is submerged in strong SAR echoes by gathering energy in the time-frequency domain. A detector called the log t-constant false alarm rate (CFAR) [29], which is based on the time-frequency amplitude spectrogram of SAR echoes, can effectively detect wideband RFIs. It is particularly useful for detecting RFIs in the Weibull background. In the same year, a method called the semantic cognitive enhancement network [30] was proposed for detecting RFIs signatures. This method can recognize RFI pixel by pixel in the SAR image domain without the need for predefined thresholds.

In the above transform domain detection methods, the flat characteristic of SAR echoes is fully utilized. When an RFI is strong, this characteristic will be destroyed so that these methods can effectively detect the pulses with RFIs. However, when the energy of an RFI is weak, the flatness cannot be exclusively relied on to effectively detect the pulses with RFIs. In addition, most of the above methods can detect whether an echo contains an RFI but cannot obtain the location of the RFI, making it easy to lose more useful signals during the suppression of RFIs, severely limiting the application range and effect of this method.

C. Solution and Contributions of This Article

In this article, a solution to the issue of detecting and locating weak energy RFIs in raw SAR data is presented. Moreover, a two-step process that utilizes the flatness and symmetry of the range spectrum to detect and locate time-varying RFIs is proposed. To improve the accuracy and reliability of RFI detection in SAR echoes, combining the flatness and symmetry indicators is recommended. This approach overcomes the limitations of using a single indicator and provides the location of RFIs. This method is more accurate and practical than traditional methods, particularly in detecting weak-energy RFIs. The results of the experiments using simulated and raw data obtained from airborne and spaceborne SAR confirm the usefulness and importance of this approach. To summarize, the main contributions of this article are as follows:

1) In this article, a new way of detecting RFIs in SAR data is presented. This method involves combining the flatness and symmetry of the SAR range spectrum to create a composite indicator. This approach is more accurate and reliable than using a single indicator and is particularly effective in detecting weak-energy RFIs. Overall, the method using the composite indicator is superior to traditional RFI detection methods in terms of accuracy and practicality.
2) A two-step approach is proposed to detect and locate time-varying RFIs more accurately. This method can precisely identify the position of RFIs in SAR echoes, even when dealing with weak energy RFIs. In contrast to conventional methods, by using a composite indicator, this method considers both the detection accuracy and localization capabilities of RFIs, resulting in more robust performance. Therefore, it has significant potential for detecting and locating RFIs and provides the basis for high-precision suppression.
3) In this article, experiments on simulated data and raw data obtained by L-band and C-band SAR are conducted to detect and locate RFIs. The results demonstrate that this method employing a composite indicator is highly efficacious in detecting and locating weak energy RFIs in echoes. This approach prevents missed RFI detection and reduces the loss of SAR signals during RFI suppression. It provides a strong and efficient solution for detecting and locating RFIs in SAR data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

D. Organization of This Article

This article is divided into several sections. In Section II, different types of RFIs are explained, and the properties of time-varying RFIs are examined in SAR data. In Section III, the composite indicator used for detecting RFIs is briefly introduced. In Section IV, a detailed explanation of the proposed method for detecting and locating time-varying RFIs is provided. In Section V, the experimental results on both simulated and measured data are presented and analyzed, followed by a discussion of potential applications of this method. Finally, in Section VI, the article is concluded.

II. Signal Model and Properties of RFI

A. Signal Model of an SAR System

The primary sources of RFIs in SAR systems are various electromagnetic devices serving within the same frequency band. In recent years, numerous instances of RFIs have been detected in SAR data, with ground communication base stations, airborne and ground-based radars, and satellites using the same frequency being identified as primary sources [31]. The forms of the RFIs caused by these interference sources to the signal of SAR include continuous, pulse, and spurious forms. Fig. 1 illustrates the spatial relationship between the irradiation area of the SAR beam and RFI sources. From the relationship between the RFI sources and the spaceborne SAR, it can be seen that ground-based sources primarily transmit RFIs to SAR through direct waves, while space-based sources can transmit RFIs through direct and scattered waves.

There are various methods to classify RFIs in an SAR system, and they can be categorized into different types based on several perspectives. In terms of the signal form, typical RFIs in SAR data can be classified as single-frequency, amplitude modulation (AM), and frequency modulation (FM) [32]. Among them, single-frequency and AM interferences are relatively narrow in bandwidth compared to SAR signals and can be classified as narrow-band interference (NBI) for RFI suppression. Moreover, FM interference, such as linear and sinusoidal frequencies, can be classified as wideband interference (WBI). Typically, RFIs are independent of SAR echoes [19], and SAR echoes that contain RFIs can be expressed as follows:

$$S(\tau, \eta) = X(\tau, \eta) + N(\tau, \eta) + R(\tau, \eta) \tag{1}$$

where $S(\tau, \eta)$ denotes the echo received by an SAR system. $X(\tau, \eta)$, $N(\tau, \eta)$, and $R(\tau, \eta)$ refer to the imaging signal, the system noise, and the RFI, respectively. $\tau = 1, 2, \ldots, N_r$ and $\eta = 1, 2, \ldots, N_a$ depict the sampling index in the range and azimuth, respectively. N_r and N_a represent the number of samples.

Although RFIs can be split into different types according to different angles, it is not conducive to the subsequent RFI suppression processing only from the perspective of the emission purpose of the interference source or the way that RFIs reach the SAR receiver. Therefore, from the perspective of the signal form, RFIs are classified into two categories, NBI and WBI, based on the bandwidth of the RFIs. The NBI signal

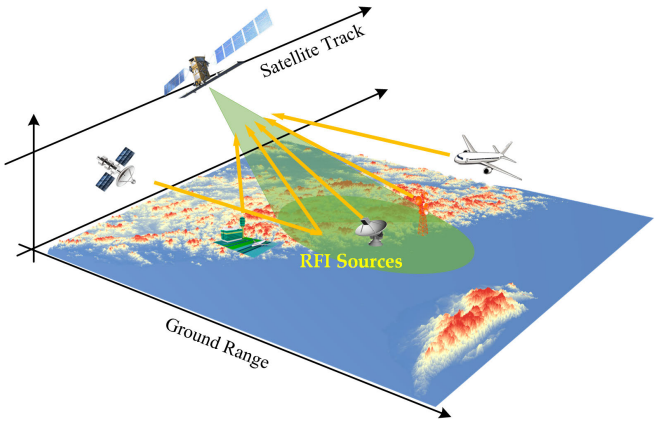


Fig. 1. Common sources of RFI and geometric model of SAR system.

model can be represented as a combination of the sinusoidal signals [33], expressed as follows:

$$R_{\mathrm{NBI}}(\tau, \eta) = \sum_{n=1}^{N} A_n(\eta) \exp(2j\pi f_n \tau + \varphi_n) \tag{2}$$

where $A_n(\tau, \eta)$ represents the amplitude, f_n represents the frequency, and φ_n represents the phase of the nth RFI signal. N is the total number of RFI signals.

Typically, the NBI signal appears as stripe-like bright lines in SAR images and has a narrow bandwidth with no complex FM. On the other hand, WBIs cause blurred artifacts to occur in SAR images due to their larger bandwidth and specific FM methods. There are two types of FM for common WBIs in SAR systems: linear FM (LFM) and sinusoidal FM (SFM) [34]. For FM interference, the corresponding signal model can be expressed as follows:

$$R_{\mathrm{WBI}}(\tau, \eta) = \sum_{n=1}^{N} A_n(\eta) \exp(j\phi) \tag{3}$$

where the phase term ϕ can be divided into LFM and SFM. These can be expressed as

$$\phi_{\mathrm{LFM}} = 2\pi f_n \tau + \pi K_n \tau^2 \tag{4}$$

$$\phi_{\mathrm{SFM}} = \beta_n \sin(2\pi f_n \tau + \varphi_n) \tag{5}$$

where K_n represents the chirp rate and β_n represents the sinusoidal modulation coefficient.

B. Characteristics of Time-Varying RFIs

In the previous research [19], a mathematical model that evaluates time-varying RFIs is established. That is, the signal-receiving window was considered in the model. The time-varying NBI is expressed as follows:

$$R_{\mathrm{TNBI}}(\tau, \eta) = \sum_{n=1}^{N} \mathrm{rect}\left(\frac{\tau}{T_d}\right) \cdot \mathrm{rect}\left(\frac{\tau - \tau_I}{T_I}\right) A_n(\eta) \cdot \exp\{j(2\pi f_n \tau + \varphi_n)\} \tag{6}$$

where T_I and τ_I refer to the pulsewidth and the central time, respectively, of the RFI received. T_d represents the duration of SAR signal reception, which can be expressed as follows:

$$T_d = \frac{2\Delta R}{c} + T_r = \frac{2(R_{\mathrm{near}} - R_{\mathrm{far}})}{c} + T_r \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

where the variables R_{near} and R_{far} represent the nearest and farthest slant ranges, respectively. c and T_r are the light velocity and pulsewidth of the signal, respectively. Two rectangular windows are used in (6) to ensure that RFIs are detected within the SAR receiving window and that they overlap in time with the SAR signal. Regarding time-varying WBIs, the same window should be considered. To facilitate the understanding of the reception of RFIs, Fig. 2 was developed to show the timing diagram of the SAR and RFI signals. In Fig. 2(a), the red and green areas are the receiving times of the RFI and SAR signals, respectively, and it has been discovered from the figure that the arrival time of RFIs falls within the receiving range of the SAR. Fig. 2(b) further describes the distribution of RFI signals in the SAR data.

Most time-varying RFIs are obtained from interference sources on the ground, and the transmitted pulses are direct waves. In this article, the one-way delay caused by the variation range is considered in the above model, and (6) is updated as follows:

$$R_{\text{TNBI}}(\tau,\eta)=\sum_{n=1}^{N}\text{rect}\left(\frac{\tau-\frac{R(\eta)}{c}}{T_d}\right)\cdot\text{rect}\left(\frac{\tau-\tau_I-\frac{R(\eta)}{c}}{T_I}\right)\times A_n(\eta)\cdot\exp\left\{j\left(2\pi f_n(\tau-\frac{R(\eta)}{c})+\varphi_n\right)\right\} \tag{8}$$

where $R(\eta)$ is the slant range model widely used in classical SAR imaging theory, which can be expressed as follows:

$$R(\eta)=\sqrt{R_0^2+V_r^2\eta^2} \tag{9}$$

where R_0 indicates the nearest slant range between the SAR and the interference source and V_r represents the equivalent velocity of the SAR platform. The same one-way delay should be taken into account by the time-varying WBIs in (3)–(5).

In preliminary studies, Li et al. [19] and [28] conducted a preliminary study on the properties of time-varying RFIs, discussing the pseudo-periodic and short-duration features of the RFIs. Fig. 3 presents an example of the data obtained from an airborne SAR system that operates in the L-band with a bandwidth of 210 MHz. In Fig. 3(a), the amplitude image in the time domain displays densely distributed bright streak artifacts, which are strong time-varying RFIs present on the SAR echoes. Fig. 3(b) highlights the region of interest (ROI) in Fig. 3(a). From the yellow rectangular area, the time-varying RFIs only account for a tiny fraction of the SAR pulse. Between the adjacent SAR pulses in the azimuth, the position of the time-varying RFI is not fixed, which is also the premise of the positioning method proposed in this article.

III. Indicators for Detecting RFIs

A. Indicator to Measure the Flatness

Generally, complex data collected by the SAR system follow a zero-mean Gaussian distribution [35]. Additionally, the amplitude of the SAR signal not polluted by RFIs is subject to the Rayleigh distribution, while the phase adheres to the uniform distribution. In the absence of RFIs, the range magnitude spectrum of the SAR echo remains relatively flat.

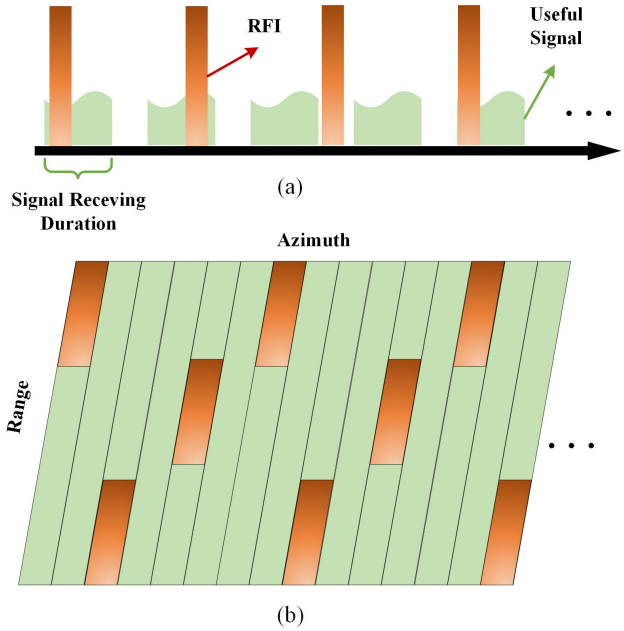


Fig. 2. Timing diagram of time-varying RFIs during SAR signal reception and its 2-D distribution in SAR data. (a) Transmitting and receiving timing diagram. (b) Two-dimensional distribution of time-varying RFIs.

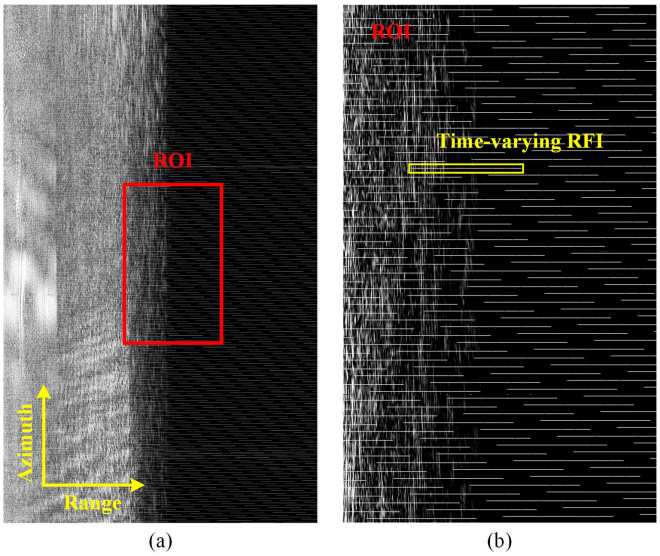


Fig. 3. Example case of the raw data obtained by airborne SAR. (a) Amplitude image in the time domain of the raw data. (b) ROI in (a).

However, when strong energy RFIs are present, the amplitude distribution becomes steeper [36], providing the basis for conventional RFI detection methods. If the RFI energy is weak, the amplitude distribution in the frequency domain may not vary enough to accurately detect its presence. At this time, performing EVD in the time domain or conducting RFI feature detection in the transform domain may result in weak energy RFIs being missed and the detection accuracy being significantly reduced.

Fig. 4 analyzes the measured data of spaceborne SAR without RFI, which comes from the Canadian RadarSat-1 payload. The data were collected on June 16, 2002, the orbit number was #34 522, and the illuminated area was the Vancouver

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

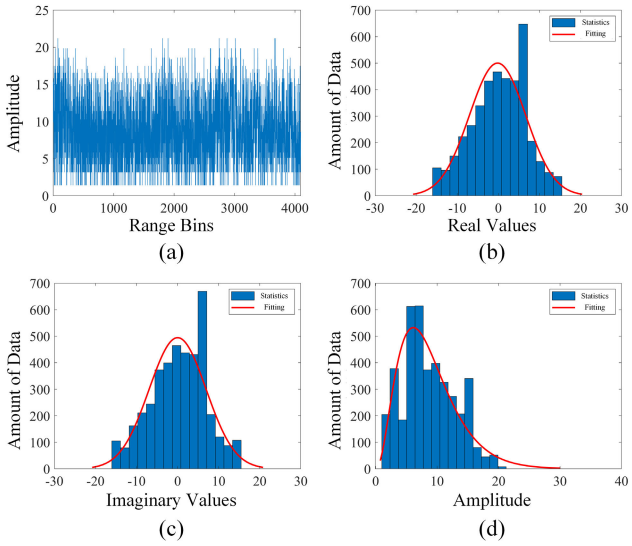


Fig. 4. Time domain characteristics of the range echo. (a) Amplitude of the echo. (b) Distribution of the echo real part. (c) Distribution of echo imaginary part. (d) Distribution of echo amplitude.

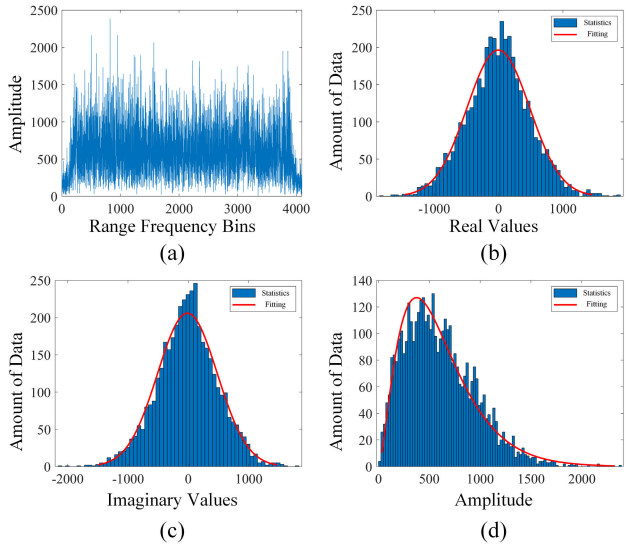


Fig. 5. Frequency domain characteristics of range echo. (a) Amplitude of the echo. (b) Distribution of echo real part. (c) Distribution of echo imaginary part. (d) Distribution of echo amplitude.

scene. Fig. 4(a) is the time-domain amplitude of an azimuth echo. Fig. 4(b) and (c) display histograms of the complex data collected by the SAR system. The red fitting curves in both images reveal that, without RFIs, both the real and imaginary parts of the SAR echo follow a Gaussian distribution. Fig. 4(d) shows a time-domain amplitude histogram of the data, where it is evident from the shape of the fitting curve that it conforms to a Rayleigh distribution.

It is assumed that a specific section of the azimuth echo signal is $S = Y + jZ$, where Y and Z represent the time-domain real part and imaginary part of the signal, respectively, and both obey the Gaussian distribution, while S obeys the Rayleigh distribution. The discrete Fourier transform of S is

$$\begin{aligned} S_{\mathrm{FT}}(k) &= \sum_{n=1}^{N} S(n) \cdot \exp\left(-\frac{j2\pi nk}{N}\right) \\ &= \sum_{n=1}^{N} (Y + jZ) \cdot \exp\left(-\frac{j2\pi nk}{N}\right) \\ &= \sum_{n=1}^{N} \left[Y \cdot \cos\left(\frac{j2\pi nk}{N}\right) + Z \cdot \sin\left(\frac{j2\pi nk}{N}\right) \right] \\ &\quad + j \sum_{n=1}^{N} \left[Y \cdot \cos\left(\frac{j2\pi nk}{N}\right) - Z \cdot \sin\left(\frac{j2\pi nk}{N}\right) \right] \\ &= Y' + jZ' \end{aligned} \tag{10}$$

where Y' is the real part and Z' is the imaginary part of the echo spectrum. Since the Fourier transform is linear, both Y' and Z' obey the Gaussian distribution, while the amplitude of S_{FT} obeys the Rayleigh distribution.

Fig. 5 presents a histogram that displays the statistical characteristics in the frequency domain of the measured data obtained by spaceborne SAR, which are similar to those shown in Fig. 4. Specifically, Fig. 5(a) shows the spectrum amplitude of the echo, while Fig. 5(b) and (c) depict histograms of the real and imaginary parts of the echo spectrum respectively. Finally, Fig. 5(d) illustrates a histogram of the echo spectrum amplitude.

B. Indicator to Measure Symmetry

Typically, the pulses transmitted and received by the radar system are real signals, and the baseband signals required for SAR signal processing are obtained through quadrature demodulation. After correcting the gain and phase of the in-phase (I) and quadrature (Q) channels, the spectrum of the signal is approximately symmetrical. This symmetry will be destroyed when RFIs exist [37]. A practical method for detecting weak RFIs is to measure the residual energy after the cancellation of spectrum energy to measure its symmetry. By combining the flatness metrics, the detection accuracy can be further improved. Based on this method, the SAR echo polluted by RFIs can be effectively detected, and the positions of the RFIs in the pulse can be obtained to provide the basis for subsequent interference suppression.

As a typical case, Fig. 6 shows the raw data of Sentinel-1 contaminated by RFIs. These data were collected on February 12, 2020. The illuminated area is Shanghai, China. There are multiple international airports and other potential interference sources in the scene. Fig. 6(a) and (b) display the magnitude images of the raw data in the 2-D time-domain and range-frequency-domain, respectively, where dense time-varying RFIs cover the SAR echoes, and some target echoes in the scene have stronger energy than the RFIs, causing the RFIs to be invisible. Fig. 6(c) and (d) show the magnitude spectrum of echoes without and with RFIs. Since the RFIs are weak, the change in the magnitude spectrum is not obvious. Therefore, the existence of weak RFIs cannot be effectively detected only by the flatness. This point is further demonstrated by the statistical distribution of the magnitude spectrum, as depicted in Fig. 6(e) and (f). Additionally, Fig. 6(g) and (h) display the energy sequence both before and after the left-right cancellation of SAR echoes. The step of left and right cancellation processing is to divide

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

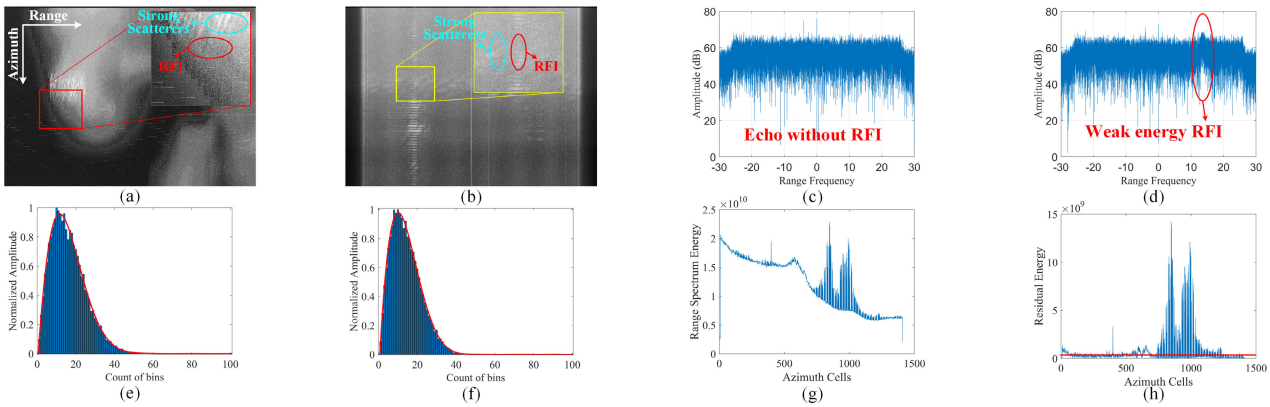


Fig. 6. Characteristics of RFIs in the measured data. (a) Amplitude image in the time domain. (b) Amplitude image in the frequency domain. (c) Magnitude spectrum of echoes without RFIs. (d) Magnitude spectrum of echoes with RFIs. (e) Amplitude histograms of echoes without RFIs. (f) Amplitude histograms of echoes with RFIs. (g) Energy sequence before the left-right cancellation of SAR echoes. (h) Energy sequence after the left-right cancellation of SAR echoes.

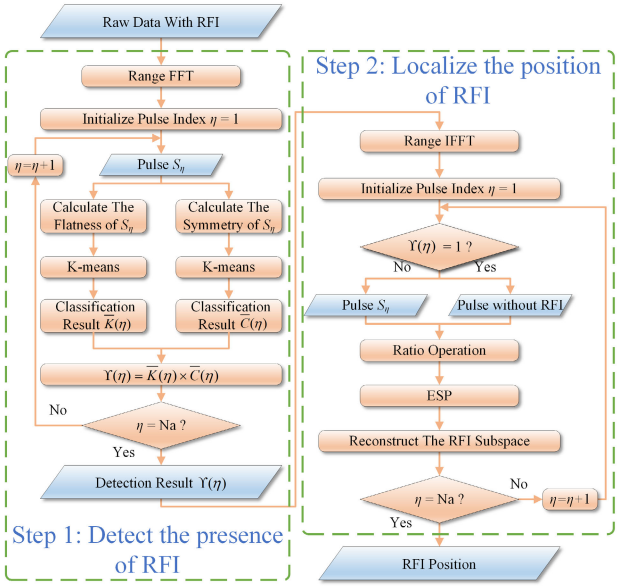


Fig. 7. Flowchart of the proposed method for detecting and localizing RFI.

the signal into two sub-bands based on the center frequency position. Then, the energy of one sub-band is subtracted from the other. Compared with the former, the residual energy sequence after cancellation is more regular, which is suitable as an indicator for detecting weak RFIs.

IV. Proposed Method

A. Detecting the Presence of RFIs in the Frequency-Domain

A method for detecting and localizing RFIs is proposed in this article to address the poor detection accuracies of traditional methods, particularly for detecting weak energy RFIs. The proposed method offers more robust detection performance and accurate localization than conventional methods. The specific process is shown in Fig. 7.

1) Calculation of the Flatness and Symmetry Indicators: To measure the instantaneous change in the signal amplitude, the concept of kurtosis is introduced into signal identification. The kurtosis is a description of the peak flatness of the data distribution. The steeper the magnitude spectrum is, the more closely the kurtosis in the frequency domain approximates the normal distribution. The formula for calculating the kurtosis in the frequency domain is

$$\kappa(\eta)=\frac{E\{(\|s_\eta\|-\mu_\eta)^4\}}{\sigma_\eta^4}=\frac{\frac{1}{N_r}\sum_{\tau=1}^{N_r}\{[\|s_\eta(\tau)\|-\mu_\eta]^4\}}{\left\{\frac{1}{N_r}\sum_{\tau=1}^{N_r}[\|s_\eta(\tau)\|-\mu_\eta]^2\right\}^2} \tag{11}$$

where s_η is the range magnitude spectrum vector, $\|\cdot\|$ is the complex amplitude, μ_η is the mean of $\|s_\eta\|$, and σ_η is the standard deviation. For a signal that approximately obeys a Gaussian distribution, when there is a mutation in the signal, $\kappa(\eta)$ will increase significantly, and its value is related to the steepness of the magnitude spectrum distribution. Therefore, the flatness of the magnitude spectrum can be used as an indicator of RFI detection.

As stated in (1), SAR data containing RFIs can be represented by $S(\tau,\eta)$. Taking into account system noise, the range energy spectrum can be calculated using FFT and Parseval's theorem as follows:

$$\mathbb{E}\{|S(f_\tau,\eta)|^2\}\approx\mathbb{E}\{|X(f_\tau,\eta)|^2\}+\mathbb{E}\{|R(f_\tau,\eta)|^2\}+\mathbb{E}\{|N(f_\tau,\eta)|^2\} \tag{12}$$

where f_τ refers to the range sampling index in the frequency domain. To perform the cancellation operation of the range energy spectrum, the signal is first divided into two sub-bands based on the center frequency position. They can be expressed as

$$\begin{aligned}c(\eta)&=\sum_{\tau=1}^{N_r}/2|S(f_\tau,\eta)|^2-\sum_{\tau=N_r/2+1}^{N_r}|S(f_\tau,\eta)|^2\\&=\sum_{\tau=1}^{N_r}/2\left(|X(f_\tau,\eta)|^2+|R(f_\tau,\eta)|^2+|N(f_\tau,\eta)|^2\right)\\&\quad-\sum_{\tau=N_r/2+1}^{N_r}\left(|X(f_\tau,\eta)|^2+|R(f_\tau,\eta)|^2+|N(f_\tau,\eta)|^2\right)\end{aligned} \tag{13}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

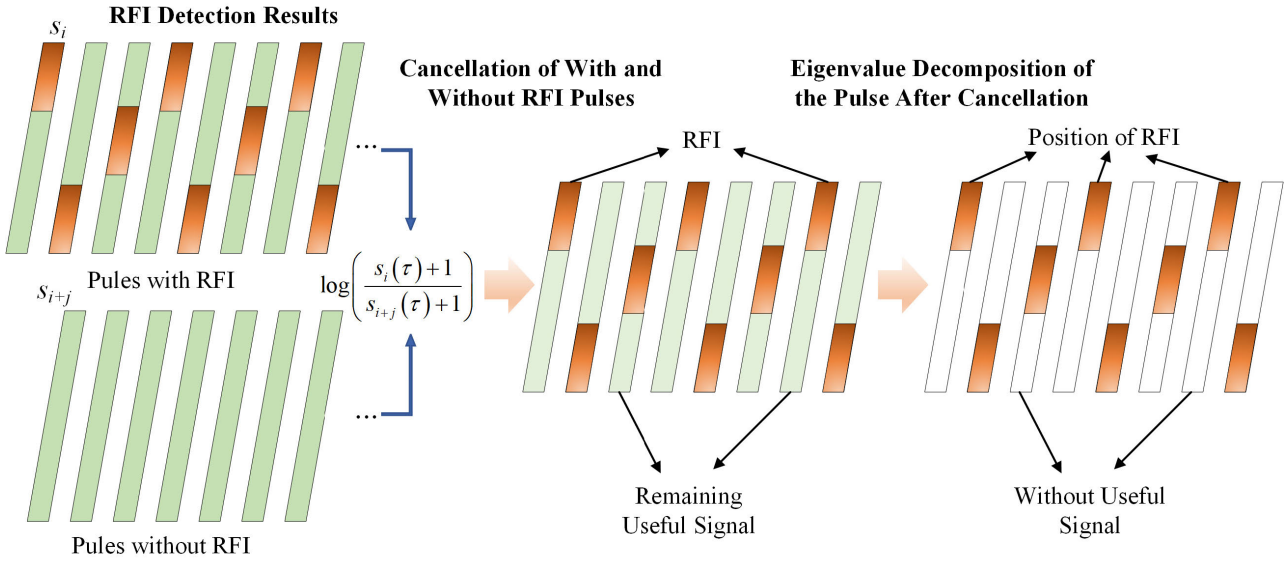


Fig. 8. Flowchart for ascertaining the positions of RFIs in the time domain.

where $c(\eta)$ stands for the remaining energy after cancellation. The energy spectrum of the imaging signal, which includes background noise, follows this equation due to the symmetrical characteristic of the range energy spectrum:

$$\sum_{\tau=1}^{N_r/2}|X(f_\tau,\eta)|^2+|N(f_\tau,\eta)|^2 \approx \sum_{\tau=\frac{N_r}{2}+1}^{N_r}|X(f_\tau,\eta)|^2+|N(f_\tau,\eta)|^2. \tag{14}$$

Then, (13) can be simplified as follows:

$$c(\eta)\approx\sum_{\tau=1}^{N_r/2}|R(f_\tau,\eta)|^2-\sum_{\tau=N_r/2+1}^{N_r}|R(f_\tau,\eta)|^2. \tag{15}$$

2) Detection of Pulses With RFIs: As mentioned in Section III, the flatness of the range magnitude spectrum and symmetry of the range energy spectrum in SAR data are disrupted by RFI. This disruption can be utilized for detection purposes. To achieve this, we first calculate the kurtosis of each echo to obtain a sequence of kurtosis values expressed as follows:

$$K(\eta)=\left[\kappa_1,\kappa_2,\ldots,\kappa_{N_a}\right]. \tag{16}$$

At this time, detecting RFIs has become a matter of evaluating the flatness and symmetry. The sequence of the kurtosis values is used to measure the flatness and is split into two categories using the K-means algorithm. One category denotes the presence of RFI signals, while the other indicates that RFIs are not detected. This process can be expressed as follows:

$$\bar{K}(\eta)=\begin{cases}0\Longleftrightarrow\text{With RFI}\\1\Longleftrightarrow\text{Otherwise.}\end{cases} \tag{17}$$

Then, the energy spectrum of all echoes undergoes left-right cancellation to derive the residual energy sequence, which can be expressed as follows:

$$C(\eta)=\left[c_1,c_2,\ldots,c_{N_a}\right]. \tag{18}$$

The K-means algorithm divides the sequence of residual energy into two categories. One category indicates the presence of RFIs, while the other denotes that no RFI has been detected. This is expressed as follows:

$$\bar{C}(\eta)=\begin{cases}0\Longleftrightarrow\text{With RFI}\\1\Longleftrightarrow\text{Otherwise.}\end{cases} \tag{19}$$

According to the previous analysis, the SAR spectrum without RFIs satisfies the flatness and symmetry conditions. Therefore, the classification results of the flatness and symmetry are comprehensively evaluated, among which those that meet the requirements of flatness and symmetry are without RFI echoes, and the rest are with RFIs, that is

$$\Upsilon(\eta)=\bar{K}(\eta)\times\bar{C}(\eta)=\begin{cases}0\Longleftrightarrow\text{With RFI}\\1\Longleftrightarrow\text{Without RFI.}\end{cases} \tag{20}$$

The detection result is represented by $\Upsilon(\eta)$, with a value of 1 indicating that RFIs are present and 0 indicating their absence.

B. Localizing the Position of RFIs in the Time-Domain

1) Cancellation of Pulses With and Without RFIs: According to the SAR imaging principle, the radar transmits coherent signals to the illuminated area at a certain pulse repetition frequency and receives the echo signal. Between adjacent pulses, the SAR travels a small distance along the track, with the ground footprint of beams remaining essentially unchanged, so there is a high correlation between adjacent pulses [38]. In addition, the existence and position of time-varying RFIs in adjacent pulses are changed. Therefore, for each pulse containing RFIs, a nearby pulse without RFIs can be selected as a reference signal for the cancellation to eliminate the imaging signal and preserve the characteristics of the RFIs. The interference subspace of the cancellation result can be constructed based on EVD, and the 1-D RFI signal is reconstructed to obtain the positioning result. The interval between the pulses with and without RFIs should not be large;

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

otherwise, the correlation between pulses will decrease, which will affect the localization results. The specific steps are shown in Fig. 8.

It is assumed that the pulses with and without RFIs are S_i and S_{i+j}. The cancellation can be depicted as follows:

$$d_i(\tau) = \log\left(\frac{s_i(\tau)+1}{s_{i+j}(\tau)+1}\right) \tag{21}$$

where the added constant 1 can prevent the numerator or denominator from becoming zero.

Then $D_i = \left[d_i(1), d_i(2), \ldots, d_i(N_r)\right]$ can be obtained after canceling all echoes containing RFIs. At this time, there are still useful signals in $d_i(\tau)$. To effectively separate the RFIs from them and obtain more accurate positioning results, it is necessary to perform EVD on $d_i(\tau)$.

2) EVD of the Pulse After Cancellation: According to the steps of the ESP in [17], the Hankel matrix containing the RFI pulse $d_i(\tau)$ is constructed as follows:

$$Q_i = \begin{bmatrix} d_i(1) & d_i(2) & \cdots & d_i(M) \\ d_i(2) & d_i(3) & \cdots & d_i(M+1) \\ \vdots & \vdots & \ddots & \vdots \\ d_i(L) & d_i(L+1) & \cdots & d_i(N_r) \end{bmatrix} \tag{22}$$

where L is the dimension of the Hankel matrix and $M = N_r + 1 - L$. The selection of L is related to the number of RFI signals in the SAR data. In actual data processing, it is generally set to 32 or 64. If L is set too small, the accuracy of RFI detection will decrease. On the other hand, if L is set too large, it will increase the matrix dimension and consequently raise the calculation amount required for EVD operation.

The covariance matrix of Q_i is processed by EVD, and it can be expressed as:

$$\hat{R} = Q_i Q_i^H = U \Lambda U^H \tag{23}$$

where $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_L)$ represents the eigenvalues of $\hat{R}$ and $U = \left[u_1, u_2, \ldots, u_L\right]$ denotes the corresponding eigenvectors. Then the RFI subspace can be constructed by the r dominant eigenvectors, and it can be depicted as follows:

$$U_r = \mathrm{span}\{u_1, u_2, \ldots, u_r\}, \quad 1 \le r \le L. \tag{24}$$

The threshold for selecting the r value can be determined by setting it according to the mean value plus the standard deviation of the L eigenvalues. Afterward, Q_i can be projected into the RFI subspace using the following operation:

$$Q_r = U_r U_r^H Q_i. \tag{25}$$

Finally, the binary-average method [39] is used to reconstruct Q_r into a 1-D signal. By processing pulse-by-pulse, the position of all RFIs can be determined. However, it should be noted that this proposed method relies on symmetry and flatness to detect RFIs. Thus, this method may not work when the bandwidths of the RFIs exceed that of the SAR and occupy the entire frequency unit. Furthermore, When the gap between pulses with and without RFI is large, the correlation between pulses will decrease, and false alarms will be generated while localizing the position of RFIs.

TABLE I
MAIN PARAMETERS IN THE SIMULATED EXPERIMENT

Main Parameters	Parameter Values
Sampling frequency	32.317 MHz
Slant range	988647 m
Pulse width	41.74 μs
Carrier frequency	5.300 GHz
Efficient velocity	7000 m/s
PRF	1256.98 Hz
Pulse bandwidth	30 MHz
Carrier frequency of RFI	5.305 GHz
Bandwidth of RFI	4 MHz
FM mode of RFI	LFM/SFM

V. EXPERIMENTAL RESULTS

In the experiment, simulated RFIs in the measured SAR data were used to verify the effectiveness of the suggested RFI detection method. Additionally, the proposed method was compared with ESP [17], ARS [24], and TDD [19]. Table I illustrates the main parameters in the simulated experiment.

A. Experimental Results and Analysis via the Simulated SAR Data

In the experiment, we utilized RadarSat-1 data with added simulated RFIs. Specifically, two types of RFIs were added to the RadarSat-1 data for the sake of generality. The performance of the various methods in detecting and locating RFIs was quantitatively analyzed by comparing their respective accuracy rates, and root mean square error (RMSE) values after RFI suppression. Notably, the proposed method demonstrated superior results in both aspects.

1) Detection Accuracy: The detection accuracy is used to assess the effectiveness of different methods in detecting whether a pulse contains an RFI in the simulated experiment. The calculation formula for this evaluation is as follows:

$$\mathrm{DA}(T_P, F_P) = \frac{T_P}{T_P + F_P} \tag{26}$$

where T_P refers to the correctly detected pulses containing RFIs and F_P indicates the number of falsely detected pulses.

In Fig. 9, the detection results of the four methods are displayed under varying signal-to-interference plus noise ratios (SINRs). When the SINR value is above 0 dB, the artifacts caused by RFI are weaker than strong point targets in the SAR image. When the SINR value reaches 10 dB and above, the artifacts caused by RFI will be mixed in the SAR image and difficult to distinguish. The second row shows the detection results, while the third row displays the positioning results. It is observed that when the SINR is -10 dB, that is, the RFI is stronger than the SAR signal, the four methods perform satisfactorily. However, the traditional method fails to detect all pulses containing RFIs when the SINR exceeds 0 dB. From the positioning results in Fig. 10, it can be more intuitively observed that the ARS, ESP, and TDD methods missed some RFIs. In contrast, the proposed method maintains good detection performance when the RFIs are hidden in strong echoes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

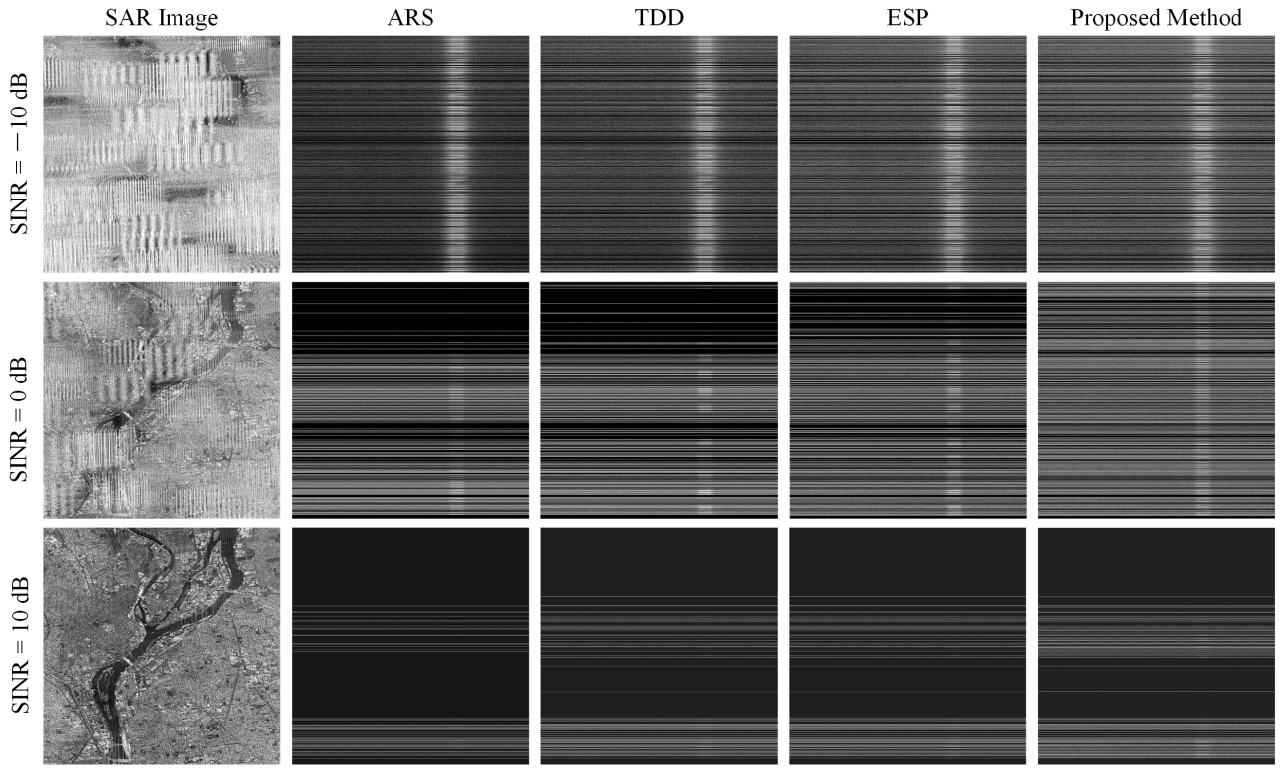


Fig. 9. RFI detection results for the ARS, TDD, ESP, and proposed method under different SINRs.

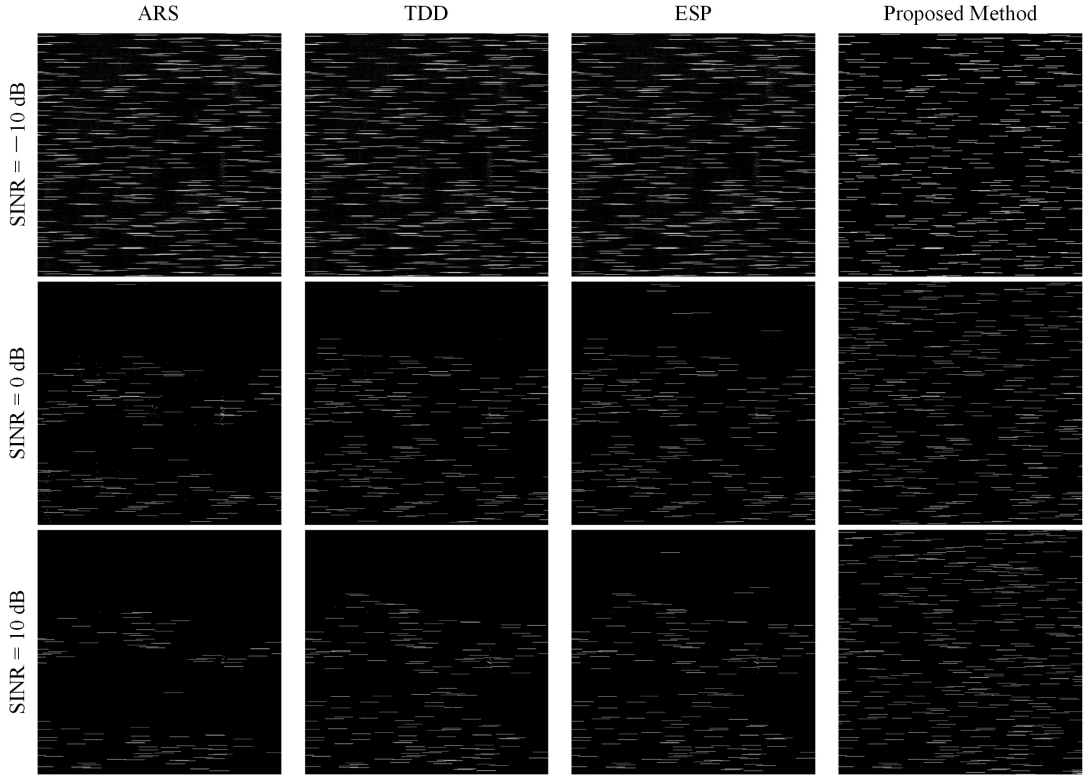


Fig. 10. RFI Localization results for the ARS, TDD, ESP, and proposed method under different SINRs.

Table II presents a comparison of the detection accuracy among the four methods. As the SINR increases, the detection performance of the traditional methods gradually deteriorates. When the SINR is 10 dB, the weak RFI signal strength makes detection more difficult, while the proposed method still achieves an exceptional detection accuracy of 97.7%. This indicates that the method utilizing the composite indicator maintains excellent performance even in scenarios with weak RFI energy in the imaging signal.

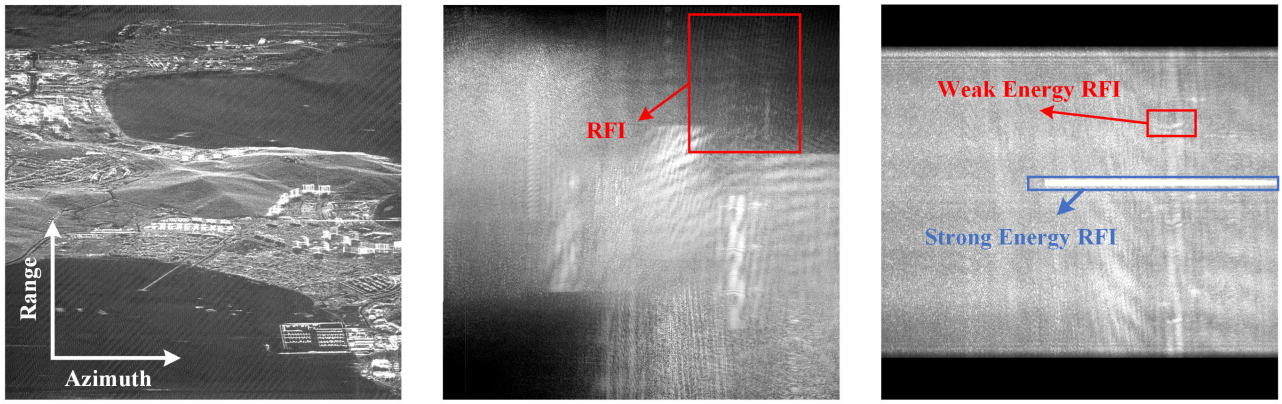
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DETECTION ACCURACY OF THE ARS, TDD, ESP, AND THE PROPOSED APPROACH

Method		ARS	TDD	ESP	Proposed Method
Detection Accuracy	SINR = −10 dB	96.72%	97.22%	97.81%	**99.25%**
	SINR = 0 dB	94.27%	95.27%	96.13%	**98.62%**
	SINR = 10 dB	90.14%	91.05%	91.44%	**97.74%**

TABLE III
RMSES AND RUNNING TIME OF FOUR METHODS

Method		ARS	TDD	ESP	Proposed Method
RMSE	SINR = −10 dB	0.2389	0.2342	0.2295	**0.2126**
	SINR = 0 dB	0.1974	0.1887	0.1851	**0.1695**
	SINR = 10 dB	0.1785	0.1759	0.1630	**0.1564**



(a) (b) (c)

Fig. 11. Experimental data of airborne SAR with RFIs. (a) Imaging result of SAR containing RFIs. (b) Amplitude image in the time domain. (c) Amplitude image in the frequency domain.

2) Performance in RFI Suppression: In addition, RFI detection is a prerequisite step for RFI suppression methods. Therefore, after using various methods to detect RFI, the same suppression method, namely FNF, was used for comparison to more intuitively demonstrate the performance of different methods in detecting RFI. In the experiment, we employed the RMSE as the evaluation index and compared the RFI suppression results of different detection methods. The calculation formula for the RMSE is as follows:

$$\mathrm{RMSE}(S, \tilde{S}) = \frac{\|S - \tilde{S}\|_F}{\|S\|_F}. \tag{27}$$

In this equation, the symbol $\|\cdot\|_F$ represents the Frobenius norm, S denotes the normalized amplitude of the non-RFI data, and the variable $\tilde{S}$ represents the normalized amplitude of the echo after RFI suppression. The RMSE can describe the magnitude of the normalized difference between the SAR data before and after RFI suppression. The smaller the RMSE value is, the more similar the echo after RFI suppression is to the original SAR data without RFIs.

Table III summarizes the comparison results of the RMSE and the average operation time of the four methods. By examining the table results, it is evident that the method utilizing the composite indicator has the lowest RMSE. However, it is less computationally efficient than the ARS and TDD methods. This shows that the proposed method suits application scenarios requiring high RFI suppression precision but low efficiency.

B. Experimental Results and Analysis via the Measured SAR Data

In the experiment, we also utilized raw data from L-band airborne SAR and C-band spaceborne SAR, which were both affected by varying degrees of RFI pollution. In this way, while verifying the performance of the proposed method in detecting weak-energy RFI, it can also be verified that the method will not generate false alarms when detecting strong-energy RFI.

Fig. 11 demonstrates the raw data of airborne SAR with RFIs. In Fig. 11(a), RFIs are clearly visible in the SAR image. Fig. 11(b) shows an amplitude image in the time domain where bright lines formed by the RFIs are densely overlaid on the SAR echo. In addition, the RFIs appear as a bright white line in the image, as can be more clearly seen from the red zoomed-in area. Fig. 11(c) is the amplitude image in the range-frequency domain of the airborne SAR raw data. Some weak energy RFIs in the red box are submerged in strong scattered echoes and are difficult to distinguish.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

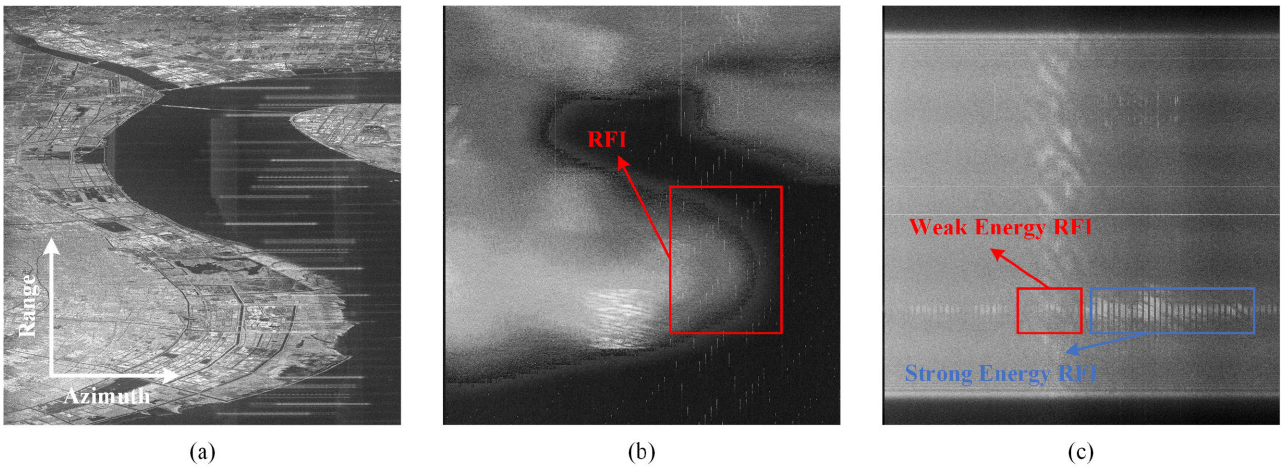


Fig. 12. Experimental data of spaceborne SAR with RFIs. (a) Imaging result of SAR containing RFIs. (b) Amplitude image in the time domain. (c) Amplitude image in the frequency domain.

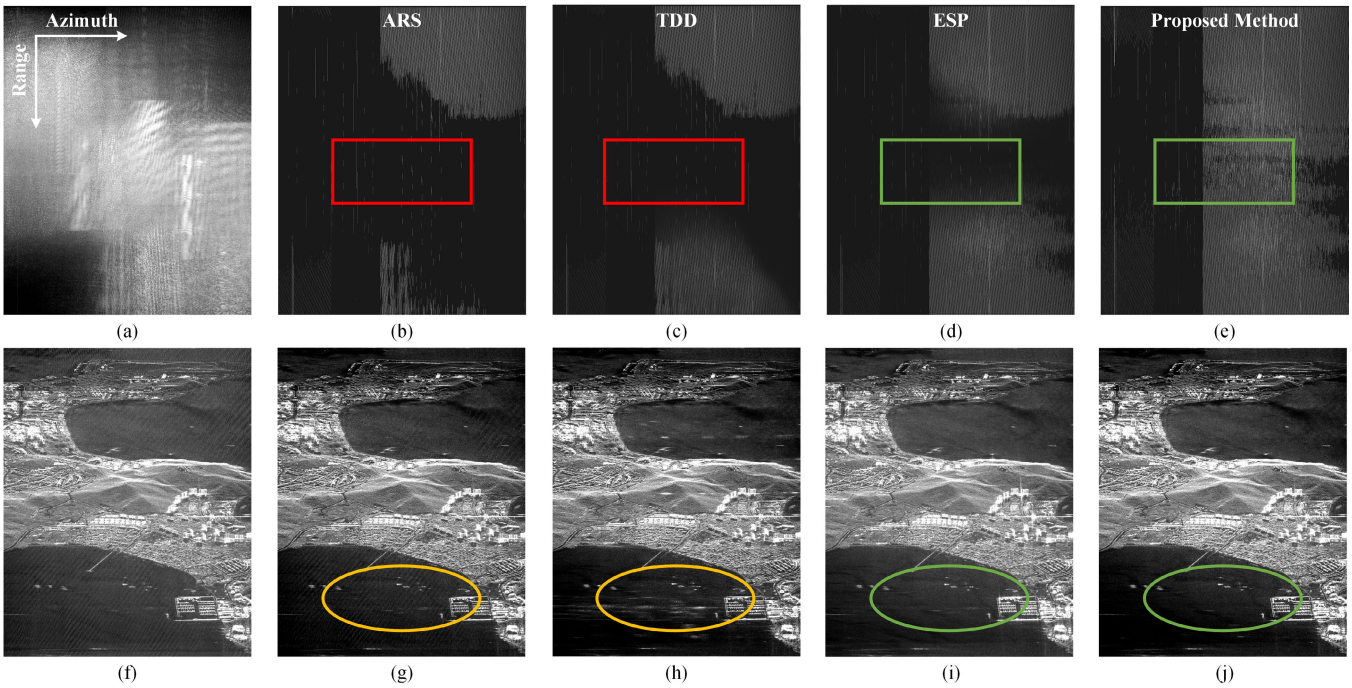


Fig. 13. Experimental results of raw data obtained by airborne SAR. (a) Amplitude image of raw data. (b) Experimental result using the ARS method. (c) Experimental result using the TDD method. (d) Experimental result using the ESP method. (e) Experimental result using the composite indicator. (f) Imaging result without mitigation process. (g)–(j) RFI mitigation results corresponding to (b)–(e), respectively.

TABLE IV
EVALUATION INDICATORS OF THE ARS, TDD, ESP, AND PROPOSED METHOD

Method	ARS	TDD	ESP	Proposed Method
Gray Entropy	0.3378	0.2864	0.2716	**0.2418**
Average Gradient	1704.9165	1727.9968	1741.7471	**1767.3715**
Running Time (s)	62.55	**54.072**	139.47	173.97

Fig. 12 shows the raw data of Sentinel-1 with RFIs. The data were collected on February 12, 2020. Fig. 12(a) is the SAR image after imaging, in which the bright strip artifacts produced by RFIs cover the ground objects. Fig. 12(b) displays the amplitude image of the raw data, where numerous RFIs are concealed within strong echoes and are challenging to differentiate. On the other hand, Fig. 12(c) illustrates the amplitude image of the data after the range FFT. As a result of weak RFI signals, changes in the echo range magnitude spectrum are not easily noticeable. Therefore, relying solely on flatness is insufficient for effectively detecting RFIs.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

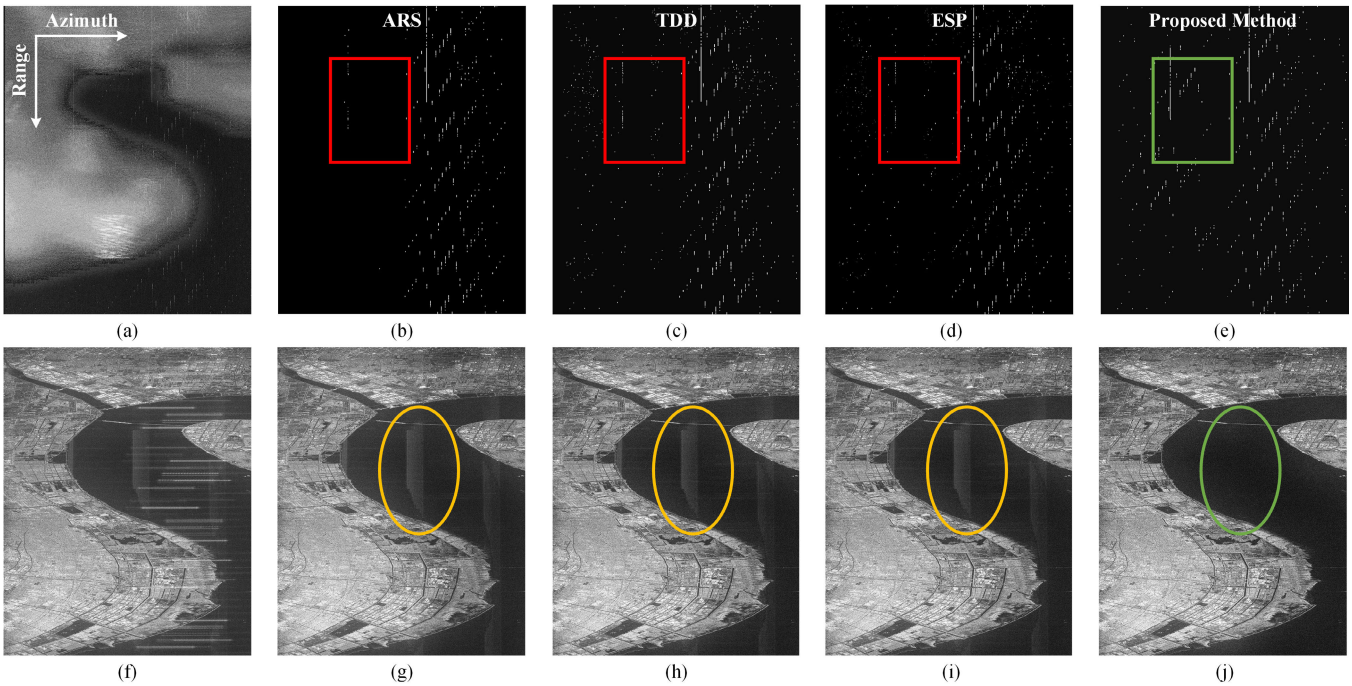


Fig. 14. Experimental results of the raw data obtained by Sentinel-1. (a) Amplitude image of the raw data. (b) Experimental result using the ARS method. (c) Experimental result using the TDD method. (d) Experimental result using the ESP method. (e) Experimental result using the composite indicator. (f) Imaging result without mitigation process. (g)–(j) RFI mitigation results corresponding to (b)–(e), respectively.

TABLE V
EVALUATION INDICATORS OF THE FOUR METHODS

Method	ARS	TDD	ESP	Proposed Method
Gray Entropy	0.7781	0.6735	0.6247	**0.5904**
Average Gradient	2492.8844	2501.7072	2537.5983	**2667.3205**
Running Time (s)	17.19	**14.86**	38.33	47.81

1) Experimental Results of Airborne SAR Data: Fig. 13 presents the experimental results obtained from airborne SAR data. Different methods were employed to detect RFIs, and the detected RFIs were suppressed using the FNF method. Fig. 13(a) displays an amplitude image of SAR raw data that contains a significant amount of RFIs. The detection outcomes of the ARS, ESP, TDD, and our proposed method are shown in Fig. 13(b)–(e), respectively. Additionally, Fig. 13(f) illustrates the imaging results without RFI detection and suppression where white linear bright bars caused by RFIs cover the entire SAR image. The four methods exhibit considerable differences in the detection performance, as depicted in Fig. 13(g)–(j). Traditional techniques fail to identify hidden RFIs within strong energy echoes, resulting in artifacts remaining even after the RFIs are suppressed. However, our proposed approach maintains good detection performance.

Two indicators, i.e., gray entropy [40] and average gradients [41], are employed to quantitatively analyze the suppression results. The gray entropy of an image is a statistical form of features that reflects the amount of information in the image. The average gradient represents the difference in the image boundary, which is the rate of change of gray levels. This value can be used to quantify the definition of an image. Assuming that p_i represents the probability of the gray value in the SAR image, the two indicators are defined as follows:

$$E = -\sum_{i=0}^{255} p_i \log p_i \tag{28}$$

$$\mathrm{AG} = \frac{\frac{1}{4}\sqrt{\left\|\frac{\partial \tilde{S}}{\partial \tau}\right\|_F^2 + \left\|\frac{\partial \tilde{S}}{\partial \eta}\right\|_F^2}}{N_a N_r}. \tag{29}$$

Table IV presents the gray entropy, average gradient, and running time of the four methods. After reviewing the results in Table IV, it is clear that the method utilizing the composite indicator outperforms the other three methods with regard to the gray entropy and average gradient. This indicates that the suggested method produces a suppression result with rich image information and clear edge outlines. Therefore, these results further confirm the efficacy and preeminence of the composite indicator.

2) Experimental Results of the Spaceborne SAR Data: The experimental results from spaceborne SAR data are presented in Fig. 14. The amplitude image of the Sentinel-1 data is shown in Fig. 14(a), which includes more weak energy RFIs. Fig. 14(b)–(e) display the detection outcomes obtained through

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

four different methods. Fig. 14(f) shows the imaging result without RFI detection and suppression processing. In these data, more RFIs were missed by traditional methods, as shown in Fig. 14(g)–(j). In contrast, the proposed method maintains good detection performance when the RFIs are hidden in the strong echo signal, making the subsequent RFI suppression more thorough with no RFI artifacts missed. The gray entropy, image gradient, and running time obtained by comparing different methods are itemized in Table V.

Through simulated and measured data experiments, comparing the detection and suppression results of different methods, it is easy to see that the proposed method considers the detection accuracy and positioning ability of RFI and has significant potential compared with traditional RFI detection methods.

C. Discussion of the Computational Complexity

In this article, the efficiency of the four methods is evaluated by calculating their running time during the experiments. The airborne SAR data size is 12 000 × 10 000, while the spaceborne SAR data size is 23 389 × 1410. Tables IV and V display the running time for each program, which was tested using MATLAB R2022a on a laptop with a 2.4-GHz CPU Intel i7-12700H and 64-GB RAM. The ARS and TDD methods have the lowest timing cost, but they miss more RFIs, resulting in poor RFI suppression. The proposed process may be time-consuming, but it can detect RFIs in SAR data effectively, and its operational efficiency is similar to that of the ESP method.

VI. Conclusion

Since traditional methods have poor detection accuracy when dealing with weak energy RFIs, the locations of the RFIs need to be considered in most of these methods. In this article, a two-step method utilizing a composite indicator for detecting and locating time-varying RFIs in raw data is presented. The approach involves analyzing the range-frequency domain characteristics of the data affected by RFIs, and a composite indicator that combines flatness and symmetry is used to obtain a more robust detection performance when dealing with weak energy RFIs. On this basis, the correlation between adjacent pulses of SAR is used, and the location of time-varying RFIs is realized by combining adjacent pulse cancellation and EVD. This method employing a composite indicator has significant potential in detecting and localizing RFIs, since both the accuracy and localization ability are taken into account. It is worth noting that this method relies on symmetry and flatness to detect RFIs and does not work when the bandwidth of the RFIs exceeds the bandwidth of the SAR and occupies the entire frequency unit. Compared to traditional detection methods, this approach has been shown to be more effective and superior through experiments on simulated RFIs and raw SAR data.

Acknowledgment

The authors would like to thank the European Space Agency for providing public access to the Sentinel-1 data, which made the comparison experiments in this article possible.

References

[1] A. Reigber et al., "Very-high-resolution airborne synthetic aperture radar imaging: Signal processing and applications," *Proc. IEEE*, vol. 101, no. 3, pp. 759–783, Mar. 2013.

[2] M. Spencer and F. Ulaby, "Spectrum issues faced by active remote sensing: Radio frequency interference and operational restrictions technical committees," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 1, pp. 40–45, Mar. 2016.

[3] Y. Huang, Z. Chen, C. Wen, J. Li, X.-G. Xia, and W. Hong, "An efficient radio frequency interference mitigation algorithm in real synthetic aperture radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224912.

[4] N. Li, H. Zhang, Z. Lv, L. Min, and Z. Guo, "Simultaneous screening and detection of RFI from massive SAR images: A case study on European Sentinel-1," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5231917.

[5] M. Tao, J. Su, Y. Huang, and L. Wang, "Mitigation of radio frequency interference in synthetic aperture radar data: Current status and future trends," *Remote Sens.*, vol. 11, no. 20, p. 2438, Oct. 2019.

[6] A. Moreira et al., "Tandem-L: A highly innovative bistatic SAR mission for global observation of dynamic processes on the Earth's surface," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 2, pp. 8–23, Jun. 2015.

[7] Y. Zhang et al., "First demonstration of multipath effects on phase synchronization scheme for LT-1," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2590–2604, Apr. 2020.

[8] Y. Zhou et al., "Very high resolution SAR imaging with DGPS-supported airborne X-band data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3605–3617, 2020.

[9] Y. Li, D. Huang, S. Xing, and X. Wang, "A review of synthetic aperture radar jamming technique," *J. Radars*, vol. 9, no. 5, pp. 753–764, Oct. 2020.

[10] M. Tao, S. Lai, J. Li, J. Su, Y. Fan, and L. Wang, "Extraction and mitigation of radio frequency interference artifacts based on time-series Sentinel-1 SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5217211.

[11] F. Zhou, M. Xing, X. Bai, G. Sun, and Z. Bao, "Narrow-band interference suppression for SAR based on complex empirical mode decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 423–427, Jul. 2009.

[12] Y. Huang, G. Liao, Z. Zhang, Y. Xiang, J. Li, and A. Nehorai, "Fast narrowband RFI suppression algorithms for SAR systems via matrix-factorization techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 250–262, Jan. 2019.

[13] Y. Huang et al., "HRWS SAR narrowband interference mitigation using low-rank recovery and image-domain sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5217914.

[14] S. Joy, L. H. Nguyen, and T. D. Tran, "Joint down-range and cross-range RFI suppression in ultra-wideband SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3136–3149, Apr. 2021.

[15] Q. Lyu et al., "SAR interference suppression algorithm based on low-rank and sparse matrix decomposition in time-frequency domain," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4008305.

[16] Y. Huang, B. Zhao, M. Tao, Z. Chen, and W. Hong, "Review of synthetic aperture radar interference suppression," *J. Radars*, vol. 9, no. 1, pp. 86–106, Feb. 2020.

[17] F. Zhou, R. Wu, M. Xing, and Z. Bao, "Eigensubspace-based filtering with application in narrow-band interference suppression for SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 1, pp. 75–79, Jan. 2007.

[18] J. Feng, H. Zheng, Y. Deng, and D. Gao, "Application of subband spectral cancellation for SAR narrow-band interference suppression," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 190–193, Mar. 2012.

[19] N. Li, Z. Lv, Z. Guo, and J. Zhao, "Time-domain notch filtering method for pulse RFI mitigation in synthetic aperture radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] H. Liu, D. Li, Y. Zhou, and T.-K. Truong, "Simultaneous radio frequency and wideband interference suppression in SAR signals via sparsity exploitation in time-frequency domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5780–5793, Oct. 2018.

[21] X. Lu, J. Yang, W. Yu, W. Su, H. Gu, and T. S. Yeo, "Enhanced LRR-based RFI suppression for SAR imaging using the common sparsity of range profiles for accurate signal recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1302–1318, Feb. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.

[22] H. Liu and D. Li, "RFI suppression based on sparse frequency estimation for SAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 63–67, Jan. 2016.
[23] S. Buckreuss and R. Horn, "E-SAR P-band SAR subsystem design and RF-interference suppression," in *Proc. Sens. Manag. Environment. IEEE Int. Geosci. Remote Sensing. Symp. (IGARSS)*, Seattle, WA, USA, Jul. 1998, pp. 6–10.
[24] Y. Luo, B. Wen, and H. Zhou, "Suppressing of radio-frequency interferences in chirp radar based on averaging range spectrum," *Wuhan Univ. J. Natural Sci. Ed.*, vol. 49, no. 5, pp. 649–652, May 2003.
[25] H.-F. Zheng, L. Yang, and J. Feng, "Research on the subband subspace filtering for narrow band interference suppression in SAR," *J. Electron. Inf. Technol.*, vol. 35, no. 12, pp. 2836–2842, Dec. 2013.
[26] R. Natsuaki, T. Motohka, M. Watanabe, M. Shimada, and S. Suzuki, "An autocorrelation-based radio frequency interference detection and removal method in azimuth-frequency domain for SAR image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5736–5751, Dec. 2017.
[27] T. Bollian, B. Osmanoglu, R. F. Rincon, S.-K. Lee, and T. E. Fatoyinbo, "Detection and geolocation of P-band radio frequency interference using EcoSAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3608–3616, Oct. 2018.
[28] N. Li, Z. Lv, and Z. Guo, "Pulse RFI mitigation in synthetic aperture radar data via a three-step approach: Location, notch, and recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5225617.
[29] J. Su, M. Xi, Y. Gong, M. Tao, Y. Fan, and L. Wang, "Time-varying wideband interference mitigation for SAR via time-frequency-pulse joint decomposition algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5118816.
[30] M. Tao et al., "Radio frequency interference signature detection in radar remote sensing image using semantic cognition enhancement network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5231714.
[31] N. Li, Z. Lv, and Z. Guo, "Observation and mitigation of mutual RFI between SAR satellites: A case study between Chinese GaoFen-3 and European Sentinel-1A," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5112819.
[32] H. Yang, K. Li, J. Li, Y. Du, and J. Yang, "BSF: Block subspace filter for removing narrowband and wideband radio interference artifacts in single-look complex SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5211916.
[33] T. Miller, L. Potter, and J. McCorkle, "RFI suppression for ultra-wideband radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 4, pp. 1142–1156, Oct. 1997.
[34] W. Xu, W. Xing, C. Fang, P. Huang, W. Tan, and Z. Gao, "RFI suppression for SAR systems based on removed spectrum iterative adaptive approach," *Remote Sens.*, vol. 12, no. 21, p. 3520, Oct. 2020.
[35] I. H. McLeod, I. G. Cumming, and M. S. Seymour, "ENVISAT ASAR data reduction: Impact on SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 2, pp. 589–602, Mar. 1998.
[36] W. Han, X. Bai, W. Fan, L. Wang, and F. Zhou, "Wideband interference suppression for SAR via instantaneous frequency estimation and regularized time-frequency filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5208612.
[37] I. G. Cumming and F. H. Wong, "Digital processing of synthetic aperture radar data," *Artech House*, vol. 1, no. 3, pp. 108–110, 2005.
[38] D. Li, M. Zhan, Z. Fang, H. Xiong, and Q. Jiang, "Parameterized wideband interference suppression for SAR imaging based on HAF," *Syst. Eng. Electron.*, vol. 39, no. 3, pp. 514–521, Mar. 2017.
[39] Z. Lv, H. Zhang, N. Li, and Z. Guo, "A two-step approach for pulse RFI detection in SAR data," in *Proc. IEEE Sensors*, Sydney, NSW, Australia, Oct./Nov. 2021, pp. 1–4.
[40] Z. Tang, Y. Deng, and H. Zheng, "RFI suppression for SAR via a dictionary-based nonconvex low-rank minimization framework and its adaptive implementation," *Remote Sens.*, vol. 14, no. 3, p. 678, Jan. 2022.
[41] W. Fan et al., "Interference mitigation for synthetic aperture radar based on deep residual network," *Remote Sens.*, vol. 11, no. 14, p. 1654, Jul. 2019.



Hengrui Zhang was born in Henan, China, in 1990. He graduated from the School of Computer Network Technology of Zhengzhou University, Zhengzhou, China, in 2010, and received the M.S. degree from the School of Computer and Information Engineering, Henan University, Kaifeng, China, in 2023. He is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering, Beijing University of Posts and Telecommunications, Beijing, China.

He is with the National Key Laboratory of Microwave Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing. His research interests include synthetic aperture radar (SAR) imaging, detection and mitigation of radio frequency interference (RFI).



Ning Li (Member, IEEE) received the B.S. degree in electronics information engineering from Northeast Forestry University, Harbin, China, in 2009, the M.S. degree in communication and information system from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2012, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences (IECAS), Beijing, China, in 2015.

In July 2015, he joined the Department of Space Microwave Remote Sensing System, IECAS, where he was an Assistant Professor. Since December 2017, he has been a Full Professor with the School of Computer and Information Engineering, Henan University, Kaifeng, China. His research interests include synthetic aperture radar (SAR) and inverse SAR imaging algorithms and autofocusing techniques, SAR polarimetric theory, and SAR image processing.

Dr. Li was a recipient of the Special Prize of President Scholarship for Postgraduate Students from the University of Chinese Academy of Sciences in 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:23:51 UTC from IEEE Xplore. Restrictions apply.